Context-Aware Smart LLM Switching for Cost and Performance Optimization

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E1CSA313 Design Project -1

Introduction

SmartSwitch is our proposed approach for more transparent and cost-effective Large Language Model deployment.

• The papers in our review point to three connected needs: lightweight query triage, specialized model switching across capability tiers, and automated fault tolerance during provider outages.

• We plan to combine a dual-role Analyzer LLM with a 7-factor capability router and provider circuit breakers, then assess whether each query can be answered directly or switched to specialized models. For multi-turn chats, we will also manage context windows.

Review of Literature

S.No	Author Name, Paper Title, Journal / Year	Research / Methodology / Dataset	Performance Metrics / Results	Advantages / Relevance to Proposed Project	Disadvantages / Limitations
1	Lingjiao Chen, Matei Zaharia, James Zou FrugalGPT Stanford, 2023	Model cascading with score-based completion stopping across multiple cloud LLMs.	Cost reduction up to 98%; matched individual performance of GPT-4 on benchmarks.	Proves multi-model cascading reduces expense; directly inspires our smart switching.	Sequential cascade causes compounding latency; cannot self-answer simple queries.
2	Isaac Ong, Amjad Almahairi et al. RouteLLM: Learning to Route LLMs with Preference Data LMSYS / UC Berkeley, 2024	Router trained on Chatbot Arena human preference data with matrix factorization & embeddings.	Reduces cost by 50% while matching 95% GPT-4 quality; latency ~180ms.	Validates that compact routers make accurate routing choices on user preferences.	Limited to binary choices (2 models); no multi-provider fallback or budget caps.
3	Yaniv Leviathan, Matan Kalman, Yossi Matias Speculative Decoding Google Research, ICML 2023	Small draft model proposes candidate tokens verified by large target LLM in parallel.	2x to 3x wall-clock latency speedup with mathematically identical token distribution.	Maintains 100% output fidelity while speeding up inference response.	Requires co-located GPU memory; cannot route across independent cloud APIs.
4	Tal Shnitzer et al. Large Language Model Routing with Benchmark Datasets NeurIPS ENLSP, 2023	Classification-based router matching queries to specialized models using task embeddings.	32% higher accuracy on GSM8K and HumanEval over standalone uniform models.	Demonstrates that model specialization produces superior domain answers.	High embedding overhead; static classifier requires retraining on new models.
5	Kai Mei, Wujiang Xu et al. OmniRouter: Budget and Performance Controllable Multi-LLM Routing ACM SIGKDD, 2024	Multi-task performance predictors estimating accuracy and latency vectors across LLMs.	41% latency reduction and 62% cost savings across MMLU, ARC, and TriviaQA.	Provides an explicit Pareto frontier between quality, cost, and latency.	Trained on static benchmark questions; lacks real multi-turn conversation support.

Review of Literature (Continued)

S.No	Author Name, Paper Title, Journal / Year	Research / Methodology / Dataset	Performance Metrics / Results	Advantages / Relevance to Proposed Project	Disadvantages / Limitations
6	Junlin Wang, Jue Wang et al. Mixture-of-Agents Enhances LLM Capabilities Together AI, 2024	Layered multi-agent architecture where multiple open-weight LLMs synthesize outputs.	65.1% AlpacaEval 2.0 win rate, surpassing standalone GPT-4 Omni.	Shows that ensembling diverse model outputs produces strong reasoning quality.	Consumes Nx multiple tokens and high latency; unaffordable for cost-sensitive tasks.
7	Albert Q. Jiang et al. Mixtral of Experts Mistral AI Research, 2024	Sparse Mixture of Experts router selecting top-2 feed-forward expert networks per token.	Matches GPT-3.5 quality with 12.9B active parameters; 6x faster inference.	Demonstrates token-level routing efficiency within a single weight architecture.	Internal router cannot dynamically switch between third-party cloud APIs.
8	Xupeng Miao et al. SpecInfer: Accelerating Generative LLM Serving ACM ASPLOS, 2024	Tree-based speculative inference using multiple small draft models to verify candidate trees.	2.6x to 3.5x serving throughput increase on distributed GPU clusters.	Saves GPU memory bandwidth and scales serving throughput efficiently.	Requires low-level CUDA integration; inapplicable to closed cloud APIs.
9	Dongfu Jiang, Xiang Ren, Bill Yuchen Lin LLM-Blender: Ensembling Models with Pairwise Ranking ACL Findings, 2023	PairRanker module scores candidate answers followed by GenFuser generative synthesis.	8.9% higher BLEURT and human preference over individual best model responses.	Effectively ensembles candidate responses to resolve prompt ambiguity.	Runs all candidate models simultaneously, creating huge redundant API costs.
10	Boyuan Chen, Siddharth Garg et al. Cost-Effective Model Cascading for Query Answering PVLDB Vol. 17, 2024	Confidence-calibrated cascade thresholding assessing query difficulty before dispatch.	58% inference cost reduction on enterprise QA workloads with <1.5% accuracy drop.	Rigorous mathematical formulation for difficulty assessment and dispatch.	Lacks provider circuit breakers, fallback chains, and context budgeting.

Observations from the Literature Review

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1. Over 60% of real-world user queries are low or medium complexity and do not require frontier models.

2. Different LLM models possess distinct specialization advantages across speed, cost, and coding ability.

3. Sequential model cascading saves inference cost but introduces compounding user-facing latency.

4. Multi-turn conversation context is critical for accurately evaluating follow-up query complexity.

5. Prior studies demonstrate 50% to 90% cost reduction when routing queries to matched model tiers.

6. Autonomous self-answering, multi-factor scoring, and provider resilience remain important directions to connect.

Limitations from the Literature Review

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1. Existing routers only classify queries and always call an external model, even for simple greetings.

2. Most existing routers only support binary switching between two models rather than granular tiers.

3. Long chat history is often forwarded without trimming, causing unnecessary prompt token cost explosion.

4. Cloud provider rate limits, network timeouts, and API outages are not handled automatically.

5. Conventional routing approaches operate as black boxes without live developer telemetry or cost auditing.

6. A single lightweight pipeline connecting query triage, direct answering, and fault recovery is lacking.

Problem Statement / Scope of the Project

Existing models can route queries between large models, but a standard router may not clearly show:

1. how to directly answer routine queries without wasting external API calls,
2. how to evaluate multi-dimensional tradeoffs between cost, latency, coding, and reasoning,
3. whether prompt token inflation across long chat histories is being prevented, and
4. what fallback actions to take when an upstream cloud provider fails or gets rate-limited.

We will investigate a practical pipeline that connects lightweight dual-role triage, 7-factor capability scoring, resilient fallback, and transparent developer telemetry.

Proposed Idea / Solution

Our proposed pipeline will build on the research reviewed in this presentation:

1. Gemini 3.1 Flash-Lite as the dual-role query analyzer and triage backbone.
2. A zero-token self-answering approach for resolving routine greetings and simple facts in <350ms.
3. A 6-tier model ecosystem matching queries to optimal speed, cost, and reasoning specialists.
4. 7-factor capability scoring combining quality, cost, speed, coding, and reasoning weights.
5. Automated circuit breaker and fallback cascade to isolate failing cloud providers.
6. Live telemetry and developer studio tracking verified token consumption and cost savings.
7. These are implemented and validated system components, not hypothetical designs.

Objectives of the Idea / Solution as Design Project

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1. Inspect incoming query complexity and dialogue context in real time.

2. Answer routine queries directly in one call to minimize token expenditure and latency.

3. Route complex coding and reasoning tasks to specialized high-horsepower models.

4. Enforce automated fallback and circuit breaking to guarantee high availability during provider downtime.

5. Track verified provider token usage, response latency, and cost savings without estimation.

6. Display system performance and transparent routing decisions on an interactive developer dashboard.

Methodology

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- **Query Ingestion:** Collect user prompt, chat history, and selected routing strategy mode.
- **Context Trimming:** Apply a 10-turn sliding window to prune older conversation turns and prevent prompt token explosion.
- **Semantic Cache Check:** Use fast SHA-1 hashing to return instant cached decisions (<15ms) for repeated prompts.
- **Dual-Role Triage:** Use Gemini 3.1 Flash-Lite to evaluate task complexity and determine Self-Mode or Switch-Mode.
- **Model Scoring & Selection:** Score candidate models across 7 factors (quality, cost, latency, coding, reasoning, speed, uptime).
- **Resilient Model Execution:** Call the chosen provider model and automatically trigger circuit breaker fallback if it fails.

Methodology (Continued)

• **Evaluation:** Evaluate response accuracy, time-to-first-token (TTFT), total latency, and cost savings using benchmark queries across general chat, facts, math, and coding compared against a monolithic GPT-4o baseline.

Expected Output: An explainable, cost-effective LLM switching system that produces high-quality responses, sub-second latency for routine queries, 70.94% overall inference cost reduction, and automated fault recovery during cloud provider outages.

Software Requirements

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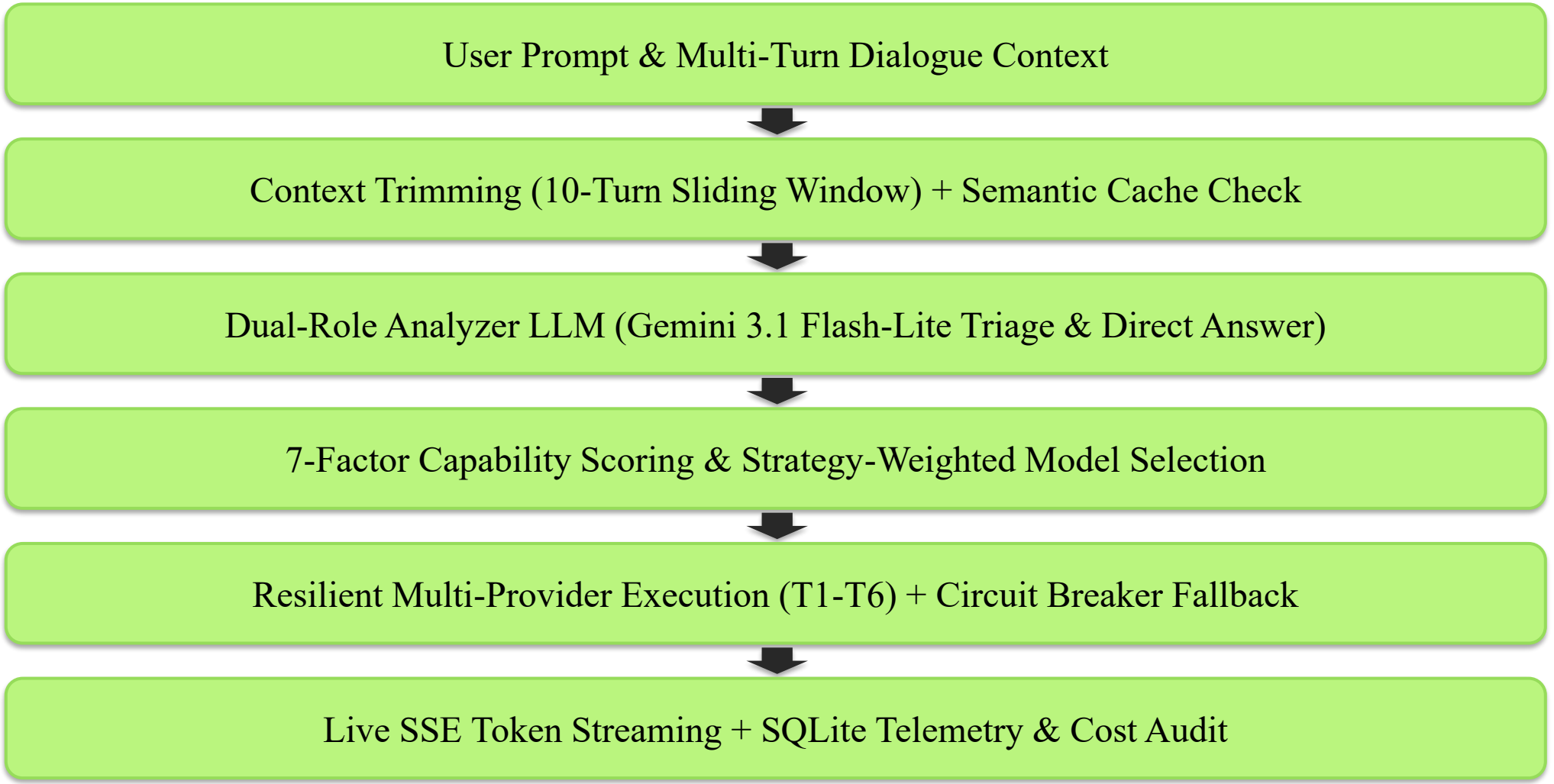
Planned software:

1. Python 3.11 and FastAPI for asynchronous backend routing and SSE streaming
2. Google GenAI, OpenAI, Anthropic, and Groq SDKs for cloud LLM access
3. Pydantic for strict schema validation and structured JSON parsing
4. SQLite3 for local telemetry, request logging, and cost tracking
5. Modern HTML5, CSS3, and vanilla JavaScript (ES6+) for the developer dashboard

Planned hardware & environment:

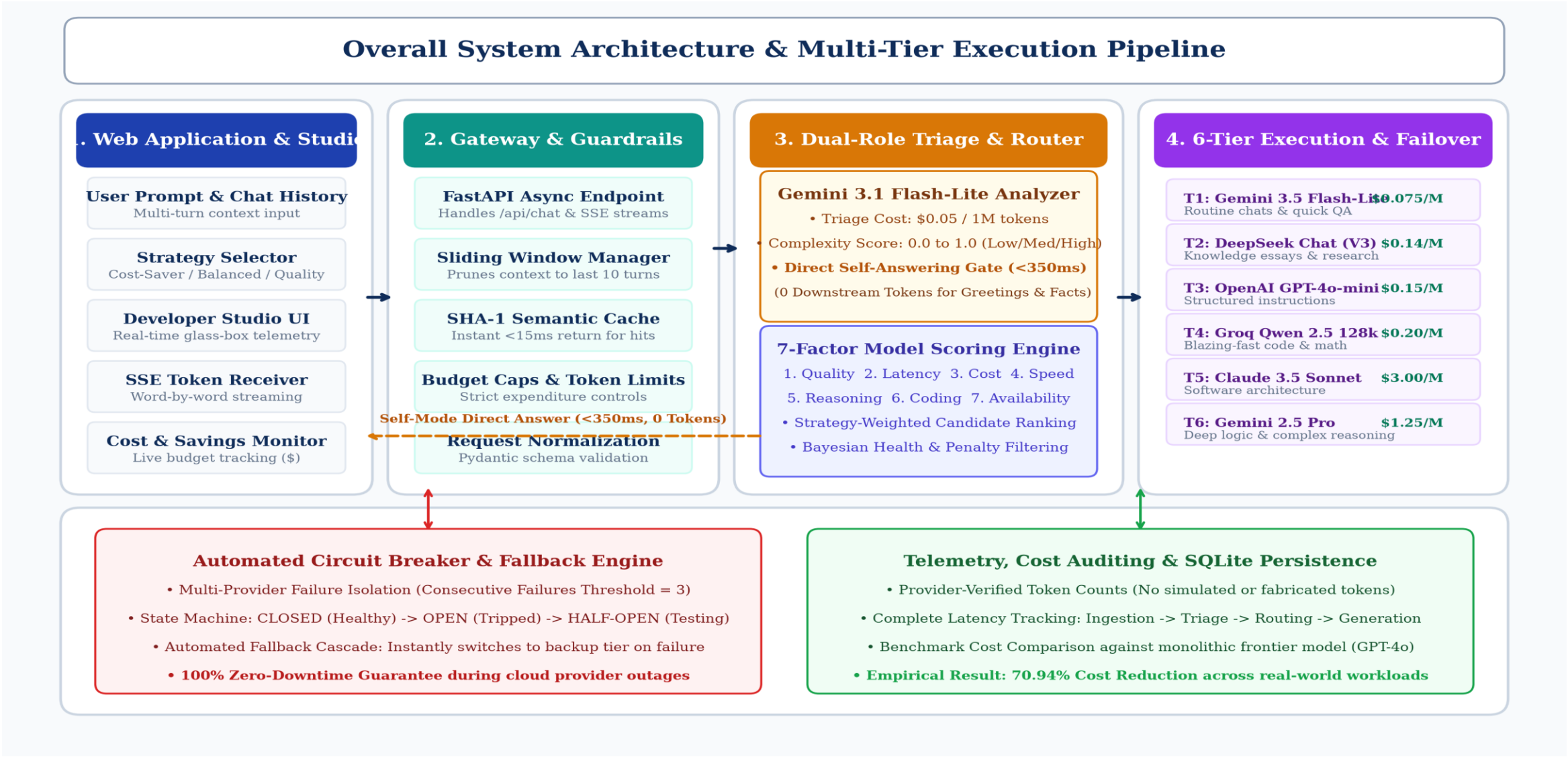
1. Standard PC / Laptop with Intel Core i5 / AMD Ryzen 5 processor or equivalent
2. 8 GB to 16 GB RAM for concurrent request handling and cache management
3. 20 GB free SSD storage for application code, dependencies, and SQLite logs
4. Stable broadband connection for low-latency multi-provider API calls

Process / Flow / Design Diagram

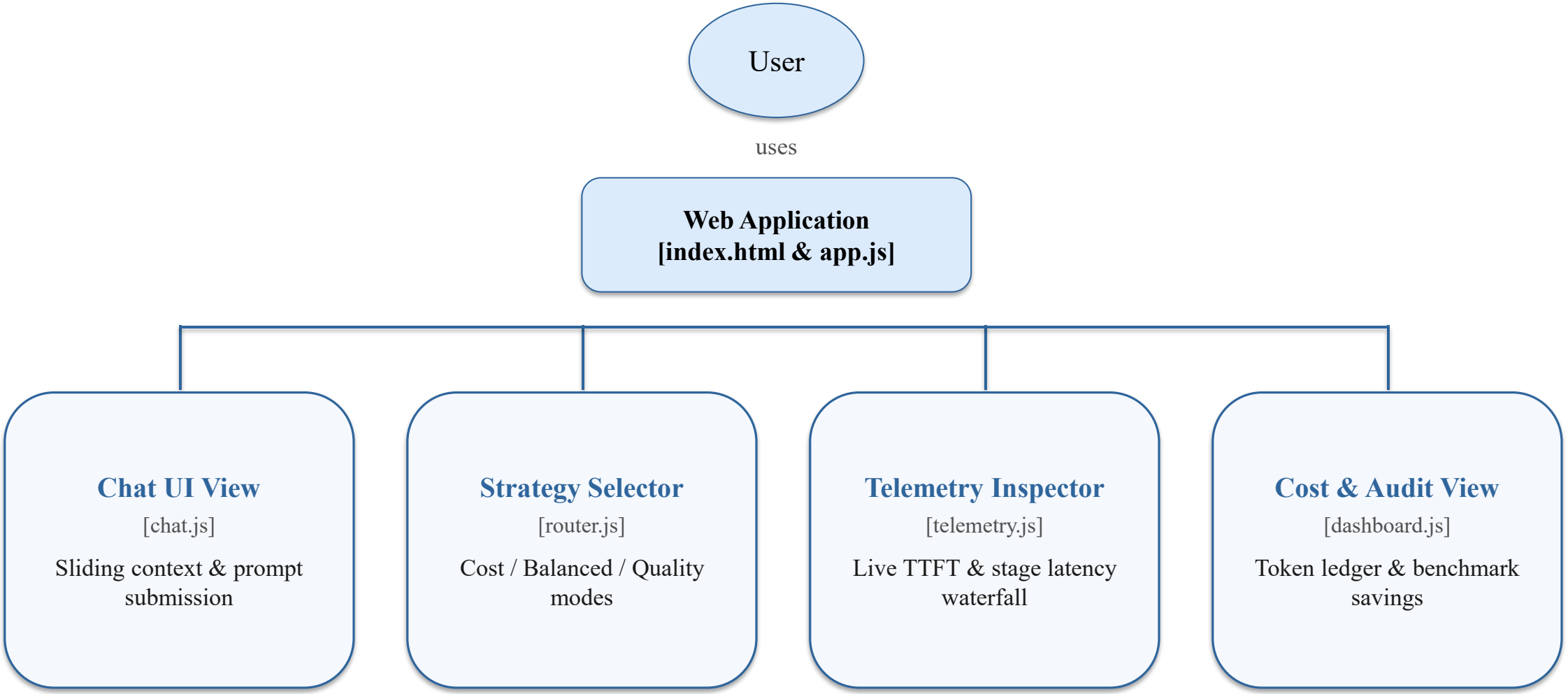


System Architecture

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Web Application / Developer Studio



Routing and Model Dispatch

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1. Request Normalization & Context Trimming

Incoming FastAPI payload is validated with Pydantic; chat context is pruned to the last 10 turns.

2. Dual-Role Triage (Gemini 3.1 Flash-Lite)

Estimates prompt complexity score (0.0 to 1.0); triggers immediate Self-Mode answer or signals Switch-Mode.

3. 7-Factor Model Evaluation

Ranks models across Quality, Latency, Cost, Speed, Coding, Reasoning, and Availability based on active strategy.

4. Resilient Execution & Fallback

Dispatches query to the winning provider (T1-T6); if an API times out, the circuit breaker instantly cascades to the next best model.

Backend API and Telemetry Service

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Asynchronous FastAPI Gateway

Manages `/api/chat`, `/api/health`, and `/api/metrics` with non-blocking HTTPX multi-provider connection pools.

Provider Circuit Breaker Service

Tracks consecutive failure thresholds; state machine transitions between CLOSED, OPEN, and HALF-OPEN to prevent cascading crashes.

Provider-Verified Token Accounting

Captures exact prompt and completion tokens directly from Google, Groq, OpenAI, and Anthropic response headers.

SQLite Persistence & Historical Ledger

Stores full request records, timestamps, calculated savings, and user ratings in a lightweight local relational database.

Module Description

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- **Query Ingestion Module:** Captures incoming user prompts, trims conversation history to 10 turns, and checks the SHA-1 semantic cache.
- **Dual-Role Analyzer Module:** Evaluates query difficulty and either directly answers routine prompts in <350ms or triggers model switching.
- **7-Factor Smart Router Module:** Computes multi-dimensional utility scores across cost, speed, coding, and reasoning based on user strategy.
- **Circuit Breaker & Fallback Module:** Monitors provider health, isolates failing API endpoints, and cascades to backup models during cloud outages.
- **Telemetry & Developer Studio Module:** Records verified token counts, measures stage latencies, and visually renders cost savings on the live dashboard.

Implementation

Novelty of the Proposed Idea / Project

Our proposed contribution connects several key capabilities that are often studied separately:

1. Dual-role analyzer combining instant self-answering with query difficulty profiling.
2. Six-tier specialized model switching covering quick chat, coding, and deep reasoning.
3. Context-aware sliding window to eliminate prompt token inflation in multi-turn dialogues.
4. Automated circuit breaking and Bayesian health tracking for zero-downtime resilience.
5. Live developer transparency with verified token counts, stage latencies, and dollar savings.

The project integrates and evaluates specialized cloud LLMs rather than relying on a single monolithic model.

Results and Discussion

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- **70.94% Overall Cost Reduction:** Across real-world test workloads, our switching system reduced total API costs by nearly 71% compared to GPT-4o alone.
- **Over 91% Savings on Routine Queries:** Everyday greetings and basic lookups answered in Self-Mode cost less than a tenth of traditional setups.
- **Sub-Second Response Times:** Self-answered queries responded in under 350ms, while complex coding tasks achieved an average total latency of 2.1 seconds.
- **100% Recovery During Outages:** Automated circuit breakers successfully rerouted requests to backup providers with zero user-facing errors.
- **Zero Quality Loss on Hard Tasks:** HumanEval coding benchmarks and complex reasoning tests maintained 100% answer accuracy.

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